

KATULA VAISHNAVI

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SUMMARY

B.Tech Data Science student (2027) with hands-on experience building end-to-end ML systems — spanning supervised learning, LLM fine-tuning, retrieval-augmented generation, and multi-agent AI. Strong foundation in model evaluation, optimization, and production-grade ML pipelines, with a focus on rigorous experimentation and measurable results.

EDUCATION

B.Tech in Data Science

Aug 2023 – Jun 2027

Malla Reddy Deemed to be University, Hyderabad, Telangana

TECHNICAL SKILLS

Languages: Python, SQL, R, C, JavaScript

ML & AI: Scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM, SHAP, HuggingFace Transformers, PEFT/QLoRA, DistilBERT, LSTM, SVM, RAGAS

Data & Tools: Pandas, NumPy, FAISS, ChromaDB, LangChain, LangGraph, Matplotlib, Seaborn

Engineering: FastAPI, Flask, Streamlit, React, Docker, Git, AWS Basics

EXPERIENCE

AI and Cloud Intern – Edunet Foundation

Jun 2025 – Aug 2025

- Built a multi-model Disease Prediction system on a clinical dataset, benchmarking Logistic Regression, Random Forest, and XGBoost classifiers for diagnostic decision support.
- Performed feature engineering, handling of missing/imbalanced data, and hyperparameter tuning, improving prediction accuracy and recall on minority disease classes.
- Evaluated models using precision, recall, F1-score, and ROC-AUC, and presented results with SHAP-based explainability to highlight key clinical risk factors.
- Deployed the trained model via a lightweight Flask API and documented the pipeline using AWS cloud fundamentals for future scalable deployment.

PROJECTS

AmazonQA-RAG

Python, QLoRA, FAISS, LangChain, RAGAS

- Fine-tuned an instruction LLM on Amazon Q&A data using QLoRA (4-bit NF4 + LoRA adapters) for efficient, low-resource adaptation.
- Built a FAISS-backed retrieval pipeline with sentence-transformer embeddings and LangChain chains for grounded, context-aware answers.
- Benchmarked Base, RAG, and Fine-tuned variants using RAGAS faithfulness and relevancy metrics, deployed via an interactive Streamlit demo.

PatientFlow AI

Python, XGBoost, LightGBM, SHAP, FastAPI

- Built a calibrated XGBoost model to predict appointment cancellation risk, improving PR-AUC by 1.84× over the base rate on 80k records.
- Forecasted patient wait times as P10/P50/P90 quantiles using LightGBM, achieving 9.1 min MAE with 79.8% empirical coverage.
- Engineered a counterfactual slot recommender and deployed the full system via FastAPI with SHAP explainability and a React frontend.

Corporate Intelligence Platform

Python, LangGraph, PyTorch, DistilBERT

- Designed a multi-agent AI platform with 5 specialised stakeholder agents orchestrated via a LangGraph supervisor into a unified briefing.
- Built an ARIMA+LSTM stock forecaster and an IsolationForest + Autoencoder anomaly ensemble for financial risk signals.
- Fine-tuned two DistilBERT classifiers for sentiment and legal text, served through a role-scoped FastAPI service with ChromaDB memory.

Matrix Factorisation Recommender

Python, PyTorch, SVD, NCF, BPR

- Implemented and benchmarked matrix factorisation models (SVD, NCF, BPR) alongside Popularity and Item-kNN baselines using a 1-vs-99 evaluation protocol with NDCG@10 and MAP@10.
- Designed a cold-start evaluation subset to assess latent-factor model performance on new users and items.
- Ran hyperparameter ablations across embedding dimension, negative sampling ratio, and dropout, and served top models via a FastAPI REST service.

CERTIFICATIONS

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| • Google GenAI AI/ML – Hack2Skill | 2025 |
| • Introduction to Technology and AWS – Coursera | 2025 |
| • Microsoft Azure AI-900 – Certiport / PearsonVUE | 2023 |
| • Infosys Pragati Cohort – SpringBoard | 2025 |
| • AWS Learning: Generative AI – AWS Skill Builder | 2026 |